Fundamentals of AI and ML Evaluated Course Project

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Project-Resume screening system

PYTHON CODE

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Welcome | resume.py 1 | resumes.csv

C: > Users > menuj > resume.py > ...

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import joblib


print("--- Loading Dataset ---")
df = pd.read_csv('resumes.csv')


def clean_function(text):
    text = re.sub('http\S+\s*', ' ', text)
    text = re.sub('#\S+', '', text)
    text = re.sub('@\S+', '  ', text)
    text = re.sub('[%s]' % re.escape("""!"#$%&'()*+,-./:;<=>?@[\]^_`{|}~"""), ' ', text)
    text = re.sub(r'[^\x00-\x7f]',r' ', text)
    text = re.sub('\s+', ' ', text)
    return text.lower()

print("--- Cleaning Resume Text ---")
df['Cleaned_Resume'] = df['Resume'].apply(lambda x: clean_function(x))


print("--- Vectorizing Text Data ---")
tfidf = TfidfVectorizer(stop_words='english', max_features=1500)
X = tfidf.fit_transform(df['Cleaned_Resume'])
y = df['Category']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

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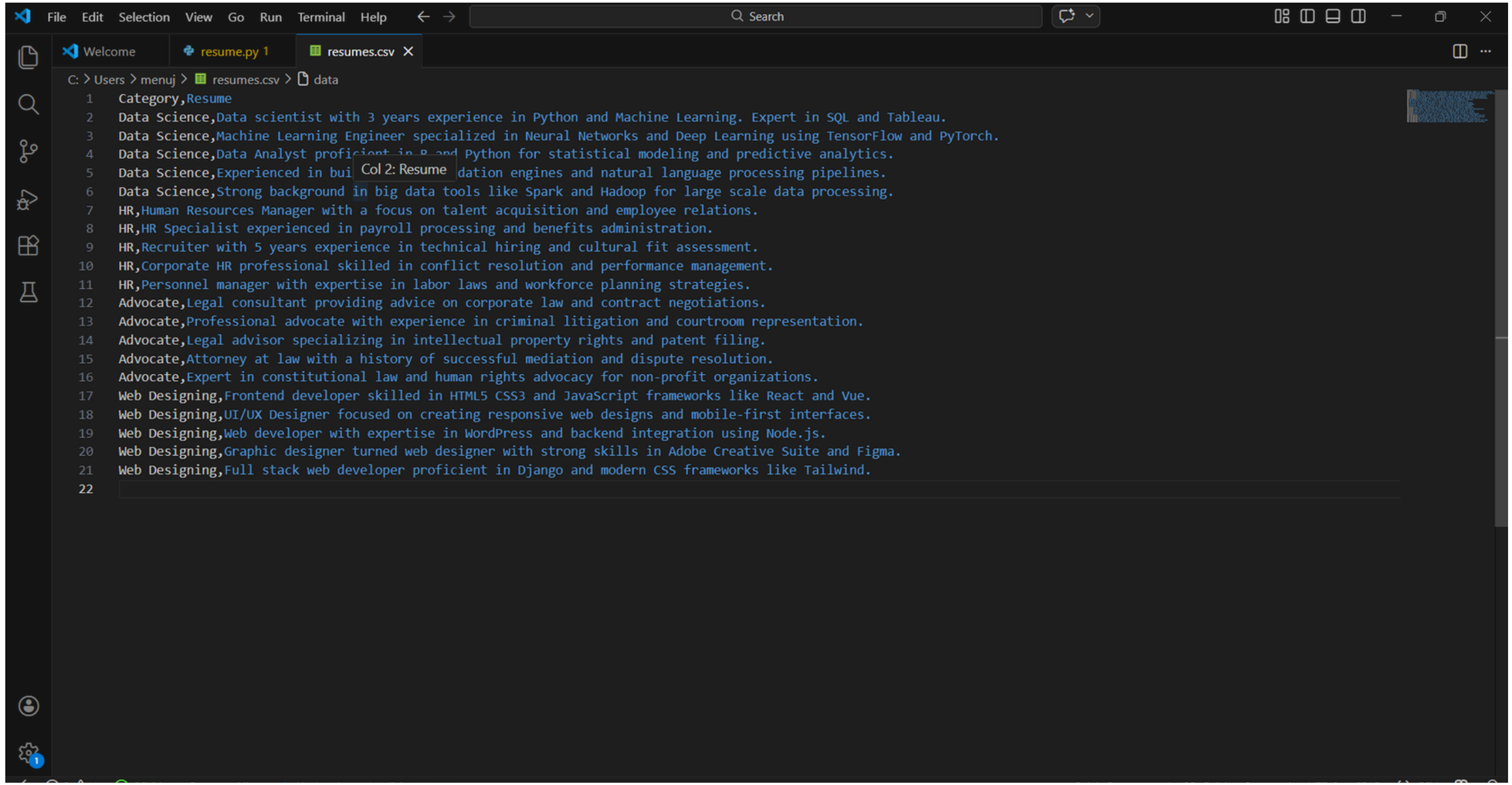
```python

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


print("--- Training Logistic Regression Model ---")
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)


y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print(f"\n[FINAL RESULT] Accuracy: {accuracy * 100:.2f}%")
print("\n--- Classification Report ---")
print(classification_report(y_test, y_pred))

plt.figure(figsize=(10, 7))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=model.classes_, yticklabels=model.classes_)
plt.title('Confusion Matrix: Resume Classification')
plt.xlabel('Predicted Category')
plt.ylabel('Actual Category')
plt.show()


print("--- Saving Model for Deployment ---")
joblib.dump(model, 'resume_model.pkl')
joblib.dump(tfidf, 'tfidf_vectorizer.pkl')
print("Files saved: resume_model.pkl and tfidf_vectorizer.pkl")
```

CSV FILE




```
Category,Resume
Data Science,Data scientist with 3 years experience in Python and Machine Learning. Expert in SQL and Tableau.
Data Science,Machine Learning Engineer specialized in Neural Networks and Deep Learning using TensorFlow and PyTorch.
Data Science,Data Analyst proficient in R and Python for statistical modeling and predictive analytics.
Data Science,Experienced in building recommendation engines and natural language processing pipelines.
Data Science,Strong background in big data tools like Spark and Hadoop for large scale data processing.
HR,Human Resources Manager with a focus on talent acquisition and employee relations.
HR,HR Specialist experienced in payroll processing and benefits administration.
HR,Recruiter with 5 years experience in technical hiring and cultural fit assessment.
HR,Corporate HR professional skilled in conflict resolution and performance management.
HR,Personnel manager with expertise in labor laws and workforce planning strategies.
Advocate,Legal consultant providing advice on corporate law and contract negotiations.
Advocate,Professional advocate with experience in criminal litigation and courtroom representation.
Advocate,Legal advisor specializing in intellectual property rights and patent filing.
Advocate,Attorney at law with a history of successful mediation and dispute resolution.
Advocate,Expert in constitutional law and human rights advocacy for non-profit organizations.
Web Designing,Frontend developer skilled in HTML5 CSS3 and JavaScript frameworks like React and Vue.
Web Designing,UI/UX Designer focused on creating responsive web designs and mobile-first interfaces.
Web Designing,Web developer with expertise in WordPress and backend integration using Node.js.
Web Designing,Graphic designer turned web designer with strong skills in Adobe Creative Suite and Figma.
Web Designing,Full stack web developer proficient in Django and modern CSS frameworks like Tailwind.
```

RESULT

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                    Python

C:\Users\menuj\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no
 predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\menuj\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Recall is ill-defined and being set to 0.0 in labels with no tr
ue samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
              precision    recall  f1-score   support

 Data Science       0.00      0.00      0.00       2.0
           HR       0.00      0.00      0.00       0.0
Web Designing       0.00      0.00      0.00       2.0

     accuracy                           0.00       4.0
    macro avg       0.00      0.00      0.00       4.0
 weighted avg       0.00      0.00      0.00       4.0

⊗ 0 ⚠ 1   Update is ready, click to restart.   Ln 65, Col 64   Spaces: 4   UTF-8   CRLF   { } Python   3.13.6   Python 3.13 (64-bit)
```

GRAPH

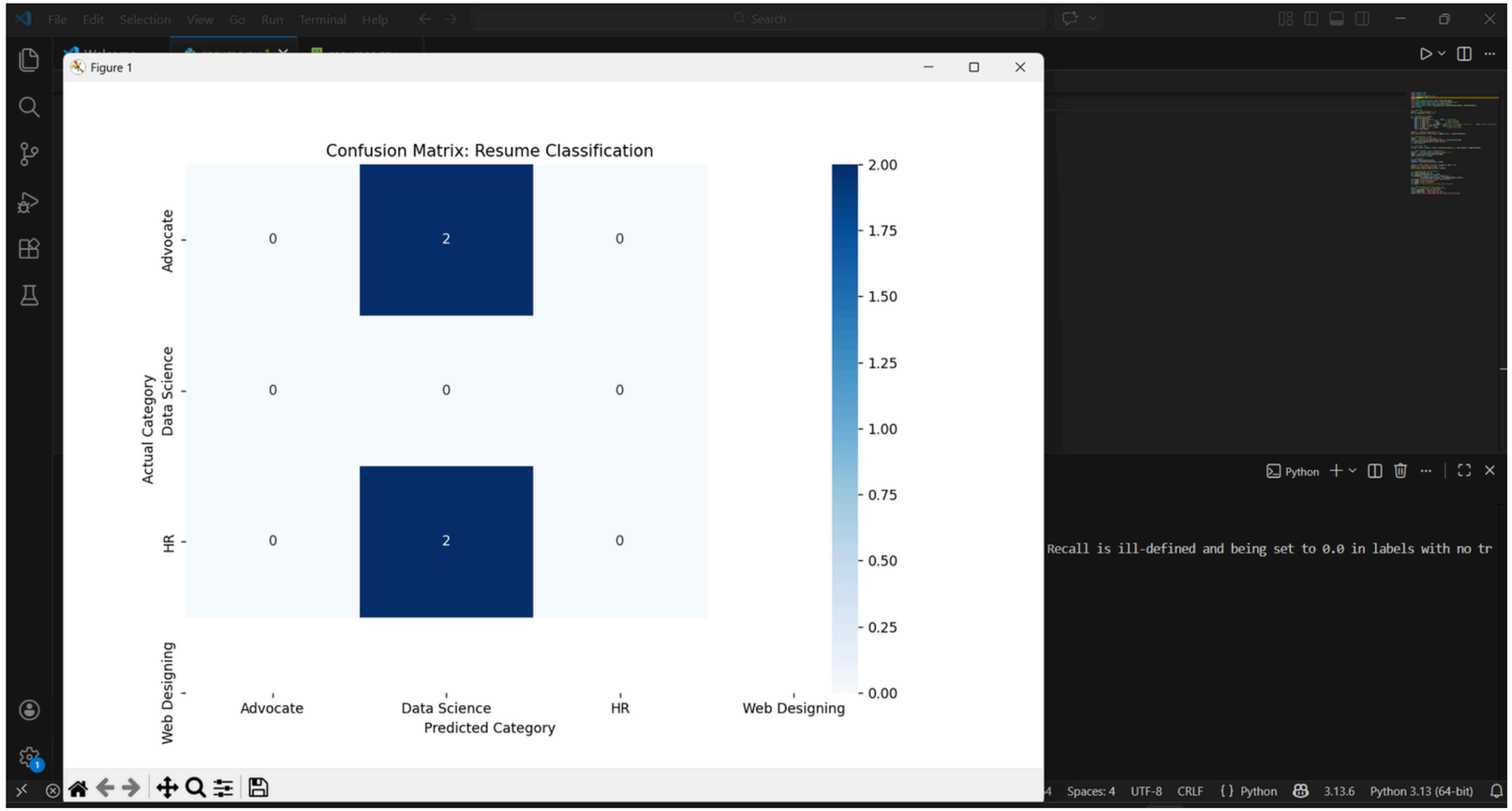


Figure 1
Confusion Matrix: Resume Classification
Actual Category
Advocate
Data Science
HR
Web Designing
Predicted Category
0
2
0
0
0
0
0
2
0
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Recall is ill-defined and being set to 0.0 in labels with no tr